

Subsystem: USU Operation System
Functional Requirement: FR-USU-1: Management of Unions

A quality requirement is a non-functional requirement that constrains how software should meet its functional requirements or how it should be developed (Pressman & Maxim).

This document defines the software quality requirements for FR-USU-1: Management of Unions in the University Student Unions system. FR-USU-1 includes key functions such as approving union membership, updating union registration data, and terminating memberships. These actions involve sensitive organisational and personal data, decision-making authority, and nationwide usage, making clear quality requirements essential to ensure the system is secure, performant, reliable, and scalable.

Each quality requirement specifies measurable and testable criteria, including a metric, unit of measurement, target value, and precision. The target value defines the desired level of performance, making the requirement falsifiable and objectively verifiable. Precision specifies the granularity of measurement, ensuring consistent and repeatable testing. This approach guarantees that all requirements are clear, unambiguous, and verifiable, enabling systematic evaluation of the system's performance, reliability, security, and scalability (Kazman; ISO/IEC 25010).

Quality Requirements

Scenario 1: Login and Authentication/ Privilege Access

Quality Attribute	Requirement statement	Metric	Unit	Precisi on	Target Value	How to Measure/ Test
Security	All USU officers must login with unique username/password (min 12 chars) + MFA, invalid attempts denied, account locks after 3 failed attempts for 20 mins.	Unauthorised access attempts.	Attempts	1	0 successful	Attempt login with invalid credentials, verify lockout occurs, check audit log.
Performance	Login response time must be fast to avoid delay.	Response Time	Secon ds	0.1	≤ 2	Automated timing of login requests under normal and peak times.
Reliability	System must correctly authenticate users on every attempt.	Authentication Success Rate	%	0.1	100	Test valid credentials multiple times, confirm system always authenticate correctly.
Scalability	System must support concurrent logins from all USU officers.	Max concurrent logins	Users	1	≥ 500	Load testing with 500+ concurrent login attempts.

Security in this context means preventing unauthorised users from accessing sensitive union data and system functions. Performance refers to how quickly the system responds to login requests. Reliability means the authentication process must work correctly every time without failure, and scalability means the system must support many officers logging in concurrently without degradation.

This scenario ensures that only authorised USU officers access FR-USU-1 functions. Security measures such as multi-factor authentication (MFA), strong password policies, session management, and audit logging create a layered defence against unauthorised access, brute-force attacks, and insider threats, ensuring compliance with GDPR. Reliability ensures the system always remains available and functional, while performance and scalability requirements guarantee smooth operation even under peak load. All aspects are measurable and testable, e.g. login response time can be timed, authentication success can be tracked, and concurrent logins can be stress-tested.

Scenario 2: Approval of Membership (FR-USU-1a)

Quality Attribute	Requirement statement	Metric	Unit	Precisi on	Target Value	How to Measure/ Test
Security	Only authenticated USU officers can approve membership; logs maintained.	Unauthorised attempts	Attempts	1	0	Attempt approval with non-officer account, check logs for completeness.
Performance	Approval processing time must be fast.	Approval processing time.	Secon ds	0.1	≤ 5	Measure time from submission to system confirmation.
Reliability	Approved memberships must be stored and accessible immediately.	Correct approvals	Count	1	100%	Approve sample memberships, verify correct data storage and retrieval.
Scalability	System must handle approval of up to 1000 memberships/hour.	Memberships processed/h our	Count	1	≥ 1000	Simulate bulk approvals under load, verify system handles expected volume.

Security in this context means only authorised USU officers can approve new unions, and all actions are fully traceable through audit logs. Performance refers to the speed at which membership approvals are processed, ensuring timely response even under heavy load. Reliability means that approved membership data is stored correctly and consistently without errors, guaranteeing correct account activation and role assignments. Scalability ensures the system can handle a high volume of membership approvals simultaneously, particularly during busy periods such as the start of the semester.

The approval process ensures that only verified unions participate in USU activities. Measures such as encryption and audit logging maintain confidentiality, integrity, and auditability, supporting accountability and compliance with data protection regulations (GDPR). All quality attributes are measurable and testable: approval processing time can be timed, correct storage of membership data can be verified, and system performance under peak load can be stress-tested to confirm scalability.

Scenario 3: Approval of Updates to USU Registration Data (FR-USU-1b)

Quality Attribute	Requirement statement	Metric	Unit	Precisi on	Target Value	How to Measure/ Test
Security	Only USU officers can approve updates, all updates logged.	Unauthorised updates	Count	1	0	Attempt update with non-officer account, verify logs capture all actions
Performance	Update approval must propagate changes quickly.	Update propagation time	Secon ds	0.1	≤ 5	Measure time from approval to data visible in system
Reliability	Approved updates must reflect correctly without errors.	Accuracy of updates	%	0.1	100	Verify updated data against intended changes in test database
Scalability	System must support 500 unions updating concurrently.	Concurrent updates handled	Count	1	≥ 500	Simulate 500 concurrent updates, confirm system stability

Security in this context means only authorised USU officers can approve updates, and all changes are fully auditable through detailed logs. Performance refers to the speed at which approved updates propagate and become visible in the system, ensuring timely access to accurate information. Reliability means that updated data must be stored correctly and consistently without errors, preserving the integrity of union registration records. Scalability ensures the system can handle multiple unions submitting updates concurrently without degradation in performance.

Scenario 3 ensures that union registration data remains accurate, secure, and up to date. Encryption protects confidentiality, and logging enables traceability and accountability, supporting compliance with data protection regulations (GDPR). All quality attributes are measurable and testable: update propagation time can be timed, data accuracy verified against intended changes, and system performance under concurrent updates can be stress-tested to confirm scalability.

Scenario 4: Approval of Termination of Membership (FR-USU-1c)

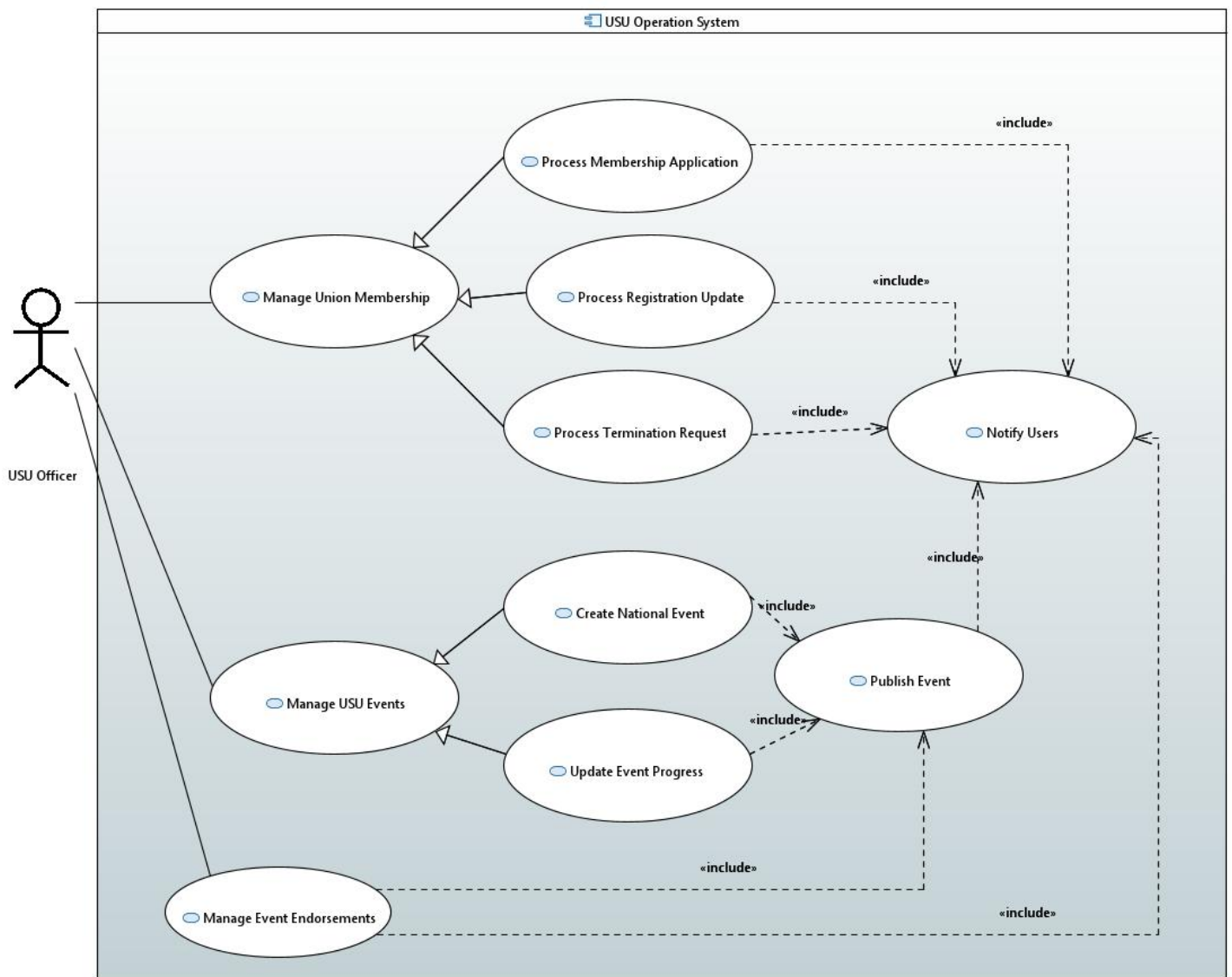
Quality Attribute	Requirement statement	Metric	Unit	Precisi on	Target Value	How to Measure/ Test
Security	Only USU officers can approve terminations, accounts locked, logs maintained.	Unauthorised termination s	Count	1	0	Attempt termination with non-officer account, check audit logs
Performance	Termination requests processed promptly.	Termination processing time	Secon ds	0.1	≤ 10	Measure time from request approval to account deactivation
Reliability	Terminated accounts deactivated, data retained 1 month then deleted.	Deactivation & deletion success	%	0.1	100	Verify deactivated accounts cannot log in, check data deletion after 1 month
Scalability	System must handle 1000 concurrent terminations.	Concurrent termination s handled	Count	1	≥ 1000	Simulate 1000 termination requests concurrently, confirm system stability

Security in this context means only authorised USU officers can approve terminations, and all termination actions are logged for traceability and accountability. Performance refers to the speed at which termination requests are processed and accounts deactivated, ensuring timely removal of inactive unions. Reliability means that terminated accounts are consistently disabled, retained securely for one month, and permanently deleted thereafter, preserving data integrity and compliance with GDPR. Scalability ensures the system can handle large volumes of terminations simultaneously without failure or performance degradation.

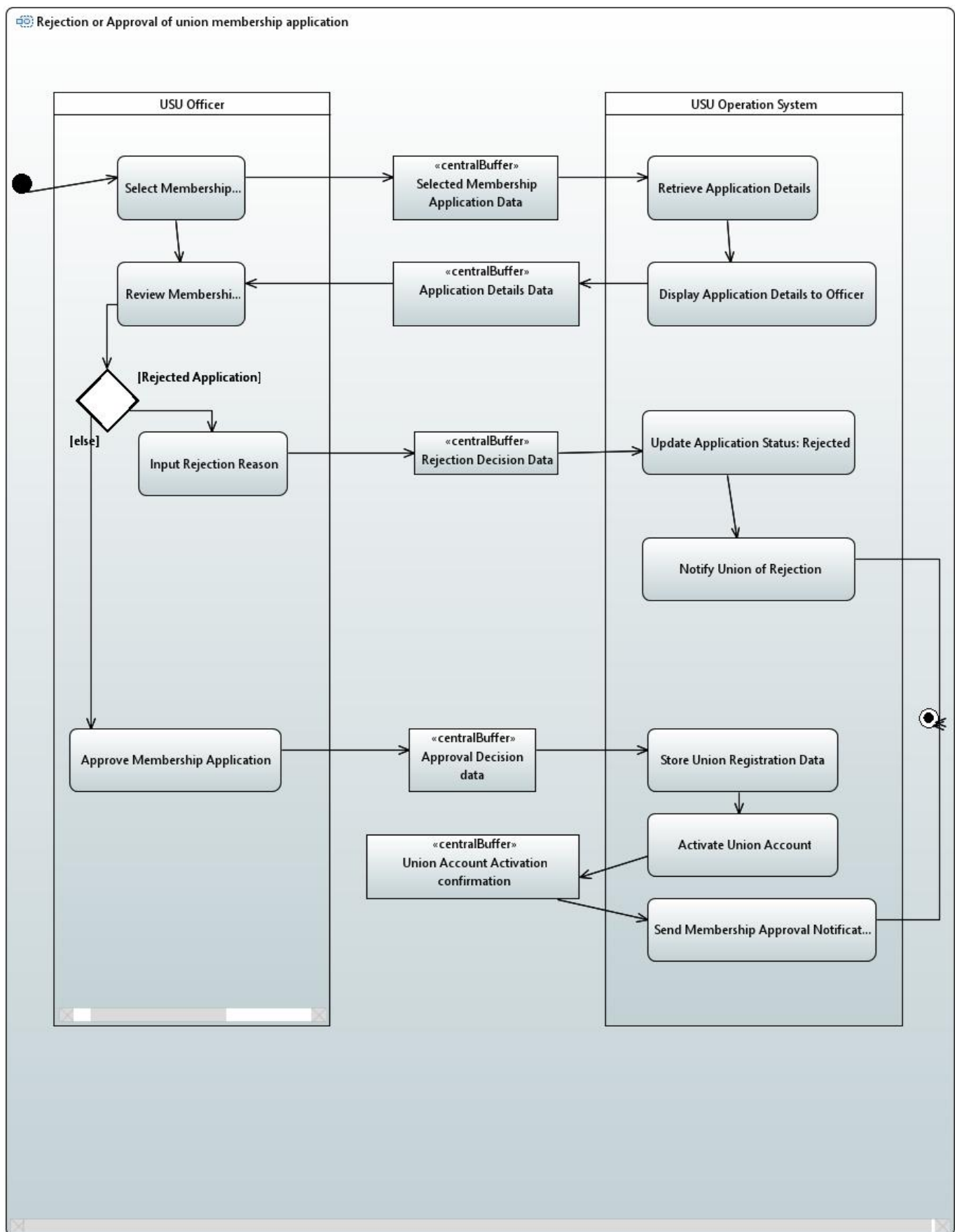
Scenario 4 guarantees that inactive unions are removed safely and securely. Temporary retention allows potential reactivation, while permanent deletion protects privacy and sensitive data. All quality attributes are measurable and testable: termination processing time can be timed, deactivation and deletion verified, and system behaviour under concurrent terminations stress-tested to confirm scalability. Logging provides a verifiable audit trail to support security and accountability.

References

Task 3a: Use Case Diagram USU-OS Subsystem

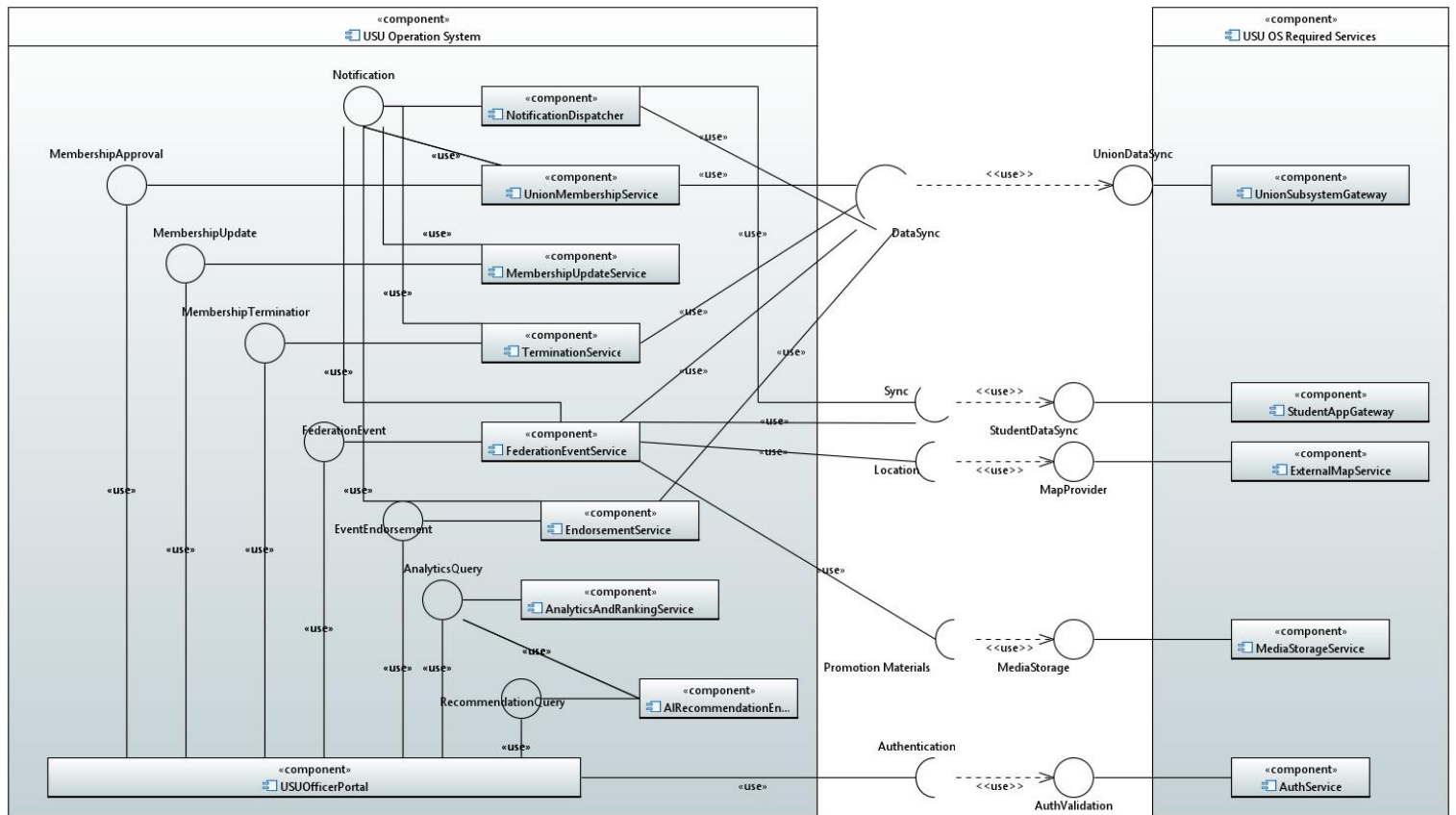


Task 3b: Activity Model: Processing Membership Application



The activity diagram models the functional requirement of processing membership applications. It covers how union officers are notified of their application outcome and the activation of union accounts if approved.

Task 4a: Architecture of the USU-OS Subsystem



Microservice and interface descriptions

UnionMembershipService

Component	UnionMembershipService
Description	Handles all membership related operations: retrieving applications from unionsubsystemgateway processing approvals or rejection decisions. Communicates decisions to external systems
Stereotype	<<microservice>>
Required Interfaces	UnionDataSync (get the applications to process), Notification
Provided Interfaces	MembershipApproval

Name	MembershipApproval	
Provider	UnionMembershipService	
Operation	Signature	<code>getApplicationDetails(in unionId: String): string</code>
	Function	Retrieves membership application data for fived union ID so officer cab review before making a decision.
Operation	Signature	<code>approveMembership(in unionid: String, in officerId: String, in decisionNote: String) : String</code>
	Function	Records an approval decision made by officer, validates the request and triggers downstream updates like account activation and notifications.
Operation	Signature	<code>rejectMembership(in unionid, String, in officerId: String, in reason: String): String</code>
	Function	Records and processes rejection decision, saves rejection record and triggers follow-up actions like notifications and audit login.

MembershipUpdate

<i>Component</i>	<i>MembershipUpdateService</i>
<i>Description</i>	Handles updates to existing union membership records and ensures relevant parties are notified.
<i>Stereotype</i>	<<microservice>>
<i>Required Interfaces</i>	Notification: used to send update alerts to relevant people.
<i>Provided Interfaces</i>	MembershipUpdate

<i>Name</i>	<i>MembershipUpdate</i>	
<i>Provider</i>	MembershipUpdateService	
<i>Operation</i>	Signature	getUpdateRequest(unionId: String): String
	Function	Retrieves pending update request for the specified union so the USU officer can review before making decision.
<i>Operation</i>	Signature	approveUpdate(in updateData: String, in officerId: String): String
	Function	Records approval of requested membership update, applies the updates details to union recird and triggers synchronisation and notification processes.
<i>Operation</i>	Signature	rejectUpdate(in updateData: String, in reason:String):String
<i>Funtion</i>		Records a rejection decision, stores officers reason and informs relevant people that the update has not been accepted.

TerminationService

<i>Component</i>	<i>TerminationService</i>
<i>Description</i>	handles process of membership termination requests submitted by unions. Retrieves termination details, applies approval decisions, deactivates union account and coordinates all required system updates.
<i>Stereotype</i>	<<microservice>>
<i>Required Interfaces</i>	UnionDataSync, Notification
<i>Provided Interfaces</i>	MembershipTermination

<i>Name</i>	<i>MembershipTermination</i>	
<i>Provider</i>	TerminationService	
<i>Operation</i>	Signature	reviewTerminationRequest(in unionId: String): String
	Function	Retrieves the termination request submitted by union, including justification and requested termination date so UO can review details.
<i>Operation</i>	Signature	approveTermination(in unionid: String, in officerId: String, in decisionNote: String): String
	Function	Records the officer's approval decision, deactivates union account, initiates the one month retention period and triggers downstream synchronisation and notification processes.

FederationEventService

<i>Component</i>	<i>FederationEventService</i>
<i>Description</i>	Manages creation, publishing and progress updates of national USU federation events. Supports fr-usu-2 and fr-uo
<i>Stereotype</i>	<<microservice>>
<i>Required Interfaces</i>	UnionDataSync, Notification, StudentDataSync, MapProvider, PromotionMaterials
<i>Provided Interfaces</i>	FederationEvent

<i>Name</i>	<i>FederationEvent</i>	
<i>Provider</i>	FederationEventService	
<i>Operation</i>	Signature	createEvent(in eventData: String, in officerId: String): String
	Function	Creates new USU federation event using the provided event details. Returns unique event identifier once the event is successfully stored.
<i>Operation</i>	Signature	publishEvent(in eventId: String): String
	Function	Publishes event across all connected subsystems distributing details to unions and students and initiating notifications.
<i>Operation</i>	Signature	updateEventProgress(in eventide: String, in progress: String): String
	Function	Updates status or progress information for an existing federation event and synchronises the update with external systems.

EventEndorsement

<i>Component</i>	<i>EndorsementService</i>
<i>Description</i>	Handles workflow for unions requesting and submitting endorsements for federations events. Supports fr-uo-3 and fr-usu-2 collaboration workflow.
<i>Stereotype</i>	<<microservice>>
<i>Required Interfaces</i>	UnionDataSync, Notification
<i>Provided Interfaces</i>	EventEndorsement

<i>Name</i>	<i>EventEndorsement</i>	
<i>Provider</i>	EndorsementService	
<i>Operation</i>	Signature	requestEndorsement(eventide: String, unionid: String): String
	Function	Creates a new endorsement request from a union for a federation event, stores the request and synchronises it through UnionDataSync. Returns a requestId string so system can track the endorsement.
<i>Operation</i>	Signature	recordEndorsement(unionid: String, eventId: String, endorsementDecision: String): String
	Function	Records the union's endorsement outcome, saves the decision details and synchronises the update via UniondDataSync. Returns confirmationId string.

NotificationDispatcher

Component	NotificationDispatcher
Description	Handles delivery of all outgoing notifications from USU-OS subsystem to unions and sydents. Supports FR notification triggers and KF-1 real time updates.
Stereotype	Service
Required Interfaces	UnionDataSync, StudentDataSync,
Provided Interfaces	Notification

Name	Notification	
Provider	NotificationDispatcher	
Operation	Signature	notifyStudents(notification: String): void
	Function	Sends event updates and relevant information to students via studentdatasync like ticket updates, event updates etc.
Operation	Signature	notifyUnionOfficers(notification: String): void
	Function	Sends targeted messages to officers (e.g. endorsement requests, membership decisions) via uniondatasync
Operation	Signature	notifiyMembershipDecision(unionid: String, decisionId: String):void
	Function	Sends membership approval, rejection or termination decisions to the appropriate union via uniondatasync.

AnalyticsAndRankingService

Component	AnalyticsAndRankingService
Description	Provides system wide and union specific analytics to support USU operational decision making. Implements key performances measurement functions (KF-analytics). Retrieves performance, engagement and global statistical data.
Stereotype	Service
Required Interfaces	UnionDataSync, StudentDataSync
Provided Interfaces	AnalyticsQuery

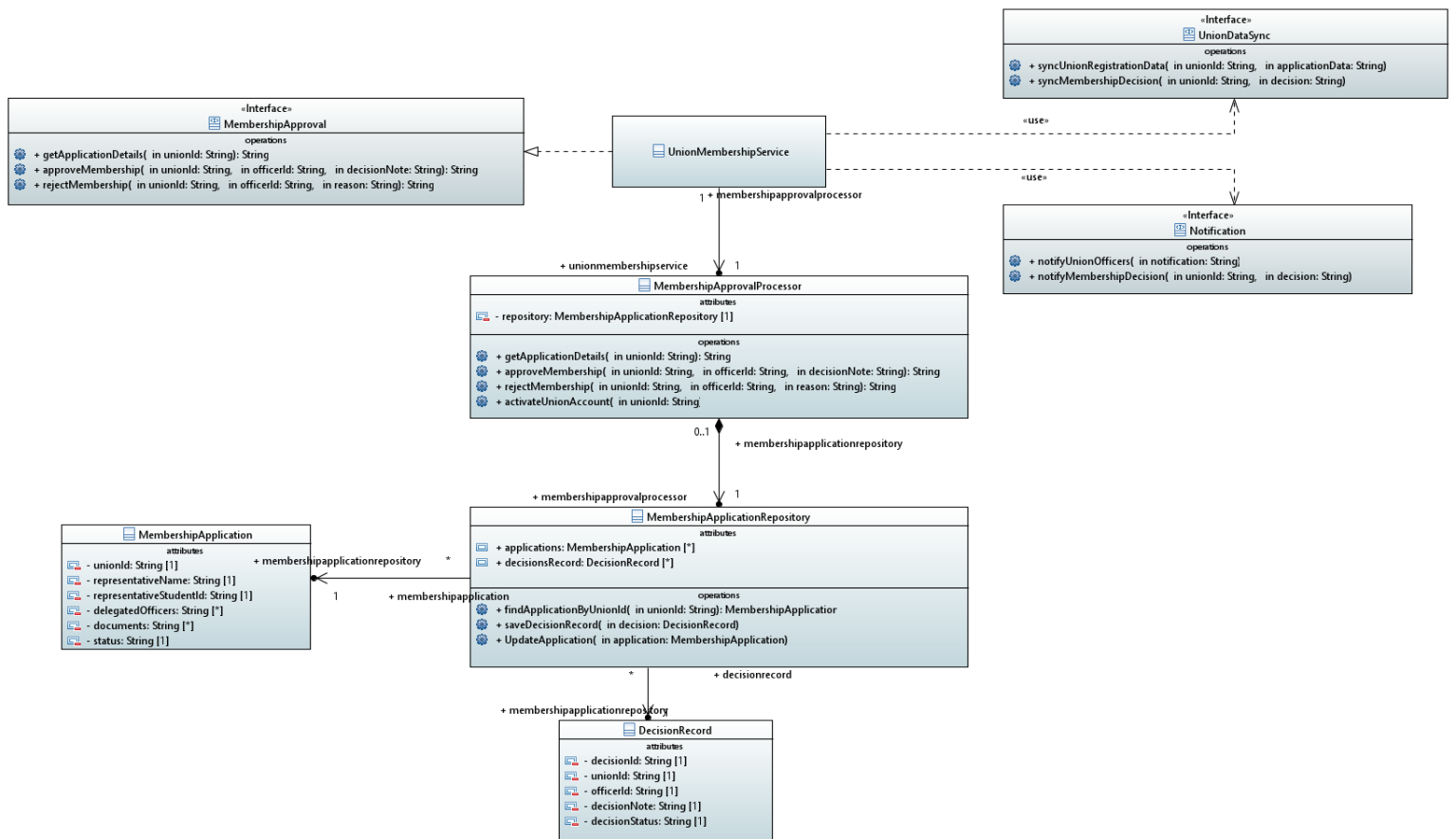
Name	AnalyticsQuery	
Provider	AnalyticsAndRankingService	
Operation	Signature	getUnionPerformance(unionid: String): String
	Function	Retrieves analytics for specific union (member activity trends, event participation, contribution metrics). Used by USU officers to assess union engagement and compliance.
Operation	Signature	getEventEngagement(eventId: String):String
	Function	Returns event specific engagement analytics (attendanace, interest rates). Supports performance evaluation of USU federation events.
Operation	Signature	getSystemStatistics():String
	Function	Provides aggregated analytics across all unions and events, supporting high-level insights required for management reporting and long-term planning.

AIRecommendationEngine

Component	AIRecommendationEngine
Description	Provides information to improve decision making for UO and USU's. Uses analytics data to suggest promotion strategies, target unions and resource allocation. Support KF-AI optimisation.
Stereotype	Service
Required Interfaces	Analytics query
Provided Interfaces	recommendationquery

Name	RecommendationQuery
Provider	AIRecommendationEngine
Operation	Signature Function
Operation	Signature Function
Operation	Signature Function

Task 5a: Structural Model



This class diagram models the internal structure of the UnionMembershipService, focusing on MembershipApproval. Attributes are encapsulated and operations match 4a ensuring consistency. Associations and multiplicities follow the logical workflow or retrieving, evaluating and persisting membership decision records.

5.b) Behavioural Diagram:

behaviour of MembershipApprovalService when it processes a rejectMembershipRequest made by USU Officer (they have approved the rejection and now the service processes it)

The sequence diagram shows the internal behaviour of the MembershipApprovalService when it processes a rejectMembership request. It uses the same microservice modelled in 5a and the steps follow the structure of the classes I defined there (MembershipApprovalProcessor, MembershipApplicationRepository and DecisionRecord).

When the service receives a rejection request it first tries to retrieve the membership application from the repository. The alt frame is used to show the 2 possible outcomes: the application is found or not. If it does not exist, the processor returns an error response.

